REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 9 Advanced | Module 3 Lesson 1 | Acceleration

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: aavg = (vf − vi)/Δt; velocity–time slope gives acceleration; area under a v–t graph gives displacement.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 54–62.

P01. Acceleration describes the rate of change of:

A. Position only B. Velocity
C. Mass D. Time

Answer

B. Velocity.

P02. The SI unit of acceleration is:

A. m/s B. m^2/s
C. m/s^2 D. s/m

Answer

C. m/s^2.

P03. A positive velocity and negative acceleration indicate:

A. Speeding up necessarily B. Negative position necessarily
C. Rest forever D. Slowing down at that instant

Answer

D. Slowing down at that instant.

P04. A horizontal velocity–time line indicates:

A. Zero acceleration B. Zero velocity necessarily
C. Increasing acceleration D. A changing displacement rate

Answer

A. Zero acceleration.

P05. The signed area under a velocity–time graph gives:

A. Acceleration B. Displacement
C. Mass D. Average force

Answer

B. Displacement.

P06. Velocity changes from +4 m/s to +16 m/s in 6 s. Calculate average acceleration.

A. 4 m/s² B. 1 m/s²
C. 2 m/s² D. -2 m/s²

Answer

C. 2 m/s². a = (vf − vi)/Δt = (16 − 4)/6 = 2 m/s².

P07. A car slows uniformly from +12 m/s to rest in 4 s. Calculate its acceleration.

A. -6 m/s² B. -1.5 m/s²
C. 3 m/s² D. -3 m/s²

Answer

D. -3 m/s². a = (0 − 12)/4 = -3 m/s², opposite the initial velocity.

P08. A velocity–time graph is horizontal at +6 m/s for 8 s. Calculate displacement from the graph.

A. 48 m B. 96 m
C. 24 m D. -48 m

Answer

A. 48 m. Displacement = area under v–t graph = rectangle = 6 × 8 = 48 m.

P09. A straight velocity–time graph rises from 0 to +12 m/s in 4 s. Calculate the displacement.

A. 48 m B. 24 m
C. 12 m D. -24 m

Answer

B. 24 m. Area under graph = ½ × base × height = ½ × 4 × 12 = 24 m.

P10. An object has initial velocity +4 m/s and constant acceleration +3 m/s² for 3 s. Find its final velocity.

A. 26 m/s B. 6.5 m/s
C. 13 m/s D. -13 m/s

Answer

C. 13 m/s. vf = vi + at = 4 + 3 × 3 = 13 m/s.

P11. Define average acceleration.

Answer

The change in velocity divided by the elapsed time: aavg = Δv/Δt.

P12. Can negative acceleration make an object speed up?

Answer

Yes. If velocity and acceleration are both negative, the speed increases while motion remains in the negative direction.

P13. What does the slope of a velocity–time graph represent?

Answer

Acceleration; positive or negative slope gives its signed direction.

P14. Distinguish velocity from acceleration.

Answer

Velocity is the rate of change of position; acceleration is the rate of change of velocity.

P15. Can an object have zero velocity and nonzero acceleration at an instant?

Answer

Yes, such as at the highest point of a vertically thrown ball while gravity still accelerates it downward.

P16. Velocity changes from +6 m/s to +24 m/s in 9 s. Calculate average acceleration.

Answer

$a = (v_f - v_i)/\Delta t = (24 - 6)/9 = 2\ m/s^2$.

P17. Velocity changes from +8 m/s to +32 m/s in 12 s. Calculate average acceleration.

Answer

$a = (v_f - v_i)/\Delta t = (32 - 8)/12 = 2\ m/s^2$.

P18. A car slows uniformly from +18 m/s to rest in 6 s. Calculate its acceleration.

Answer

$a = (0 - 18)/6 = -3\ m/s^2$, opposite the initial velocity.

P19. A car slows uniformly from +24 m/s to rest in 8 s. Calculate its acceleration.

Answer

$a = (0 - 24)/8 = -3\ m/s^2$, opposite the initial velocity.

P20. A velocity–time graph is horizontal at +9 m/s for 12 s. Calculate displacement from the graph.

Answer

Displacement = area under v–t graph = rectangle = 9 × 12 = 108 m.

P21. A velocity–time graph is horizontal at +12 m/s for 16 s. Calculate displacement from the graph.

Answer

Displacement = area under v–t graph = rectangle = 12 × 16 = 192 m.

P22. A straight velocity–time graph rises from 0 to +18 m/s in 6 s. Calculate the displacement.

Answer

Area under graph = ½ × base × height = ½ × 6 × 18 = 54 m.

P23. A straight velocity–time graph rises from 0 to +24 m/s in 8 s. Calculate the displacement.

Answer

Area under graph = ½ × base × height = ½ × 8 × 24 = 96 m.

P24. An object has initial velocity +6 m/s and constant acceleration +3 m/s² for 4 s. Find its final velocity.

Answer

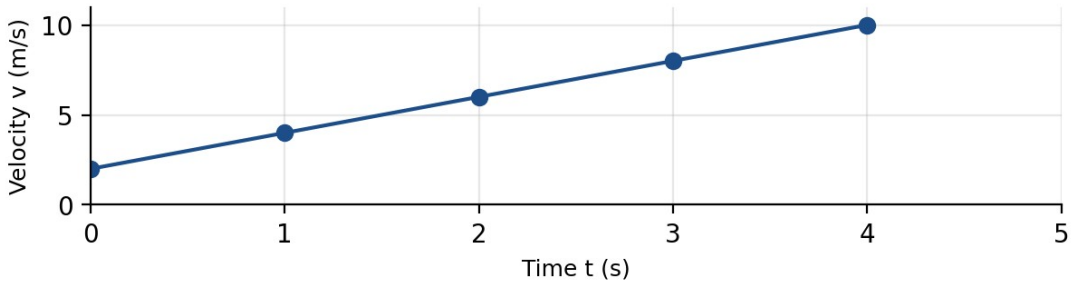
$v_f = v_i + at = 6 + 3 \times 4 = 18\ m/s$.

P25. An object has initial velocity +8 m/s and constant acceleration +3 m/s² for 5 s. Find its final velocity.

Answer

$v_f = v_i + at = 8 + 3 \times 5 = 23\ m/s$.

P26. Plot t = 0, 1, 2, 3, 4 s against v = 2, 4, 6, 8, 10 m/s. Find acceleration and displacement in 4 s.



Answer

The v–t graph is a straight line with slope (10 − 2)/4 = 2 m/s². Displacement is trapezoid area = (2 + 10) × 4/2 = 24 m.

P27. An object has v = −6 m/s and a = +2 m/s². Is it speeding up or slowing down initially? Explain.

Answer

It is slowing down because acceleration is opposite to velocity. Its velocity becomes less negative until reaching zero, after which it can move positively.

P28. A student says negative acceleration always means slowing down. Give a counterexample.

Answer

An object with v = −3 m/s and a = −2 m/s² becomes more negative in velocity; its speed increases.

P29. A v–t graph lies below the time axis. Explain the sign of its area and distinguish displacement from distance.

Answer

The signed area is negative, giving negative displacement. Distance travelled uses the magnitude of area, adding magnitudes if the graph crosses the axis.

P30. Describe two ways that velocity can change.

Answer

Its magnitude can change, its direction can change, or both can change. Any change in velocity means acceleration.